

Optimization for Learning Exercises

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Linear algebra

Exercise 1 Verify the Schur complement formula.

Derivatives with respect to matrices

Exercise 2 Let $X : \mathbf{R} \rightarrow \mathbf{R}^{m \times n}$ and $Y : \mathbf{R} \rightarrow \mathbf{R}^{n \times p}$. Show that

$$\frac{d}{dt}(X(t)Y(t)) = \frac{dX(t)}{dt}Y(t) + X(t)\frac{dY(t)}{dt}$$

Exercise 3 Show that

$$\frac{\partial}{\partial X} \operatorname{tr} AX = A^T$$

where $X \in \mathbf{R}^{m \times n}$ and $A \in \mathbf{R}^{n \times m}$

Exercise 4 Show that

$$\frac{\partial}{\partial X} \ln \det X = X^{-T}$$

where $X \in \mathbf{R}^{m \times m}$ and $\det X > 0$. Hint: Remember Jacobi's formula.

Exercise 5 Show that

$$\frac{\partial}{\partial X} a^T X^{-1} b = -X^{-T} a b^T X^{-T}$$

where $X \in \mathbf{R}^{m \times m}$ and $a, b \in \mathbf{R}^m$. Hint: First use the identity $XX^{-1} = I$ to show that

$$\frac{\partial}{\partial X_{i,j}} X^{-1} = -X^{-1} \frac{\partial X}{\partial X_{i,j}} X^{-1}, \quad i, j \in \mathbf{N}_m$$

Exercise 6 Show that

$$\frac{\partial}{\partial X} \operatorname{tr} X^T A X = (A + A^T)X$$

where $X \in \mathbf{R}^{m \times n}$ and $A \in \mathbf{R}^{m \times m}$.

Convexity

Exercise 7 Let $f : \mathbf{R} \rightarrow \mathbf{R}$ and $g : \mathbf{R} \rightarrow \mathbf{R}$ be convex functions. Show that $f + g$ is also a convex function.

Exercise 8

1. Show that $f(x) = x \ln x$ with $\mathbf{dom} f = \mathbf{R}_+$ is a convex function.
2. Show that $g(x) = \sum_{i=1}^n x_i \ln x_i$ with $\mathbf{dom} g = \mathbf{R}_+^n$ is a convex function.
3. Derive the conjugate function of f above.

Exercise 9 Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ and define $\mathbf{epi} f = \{(x, t) \mid x \in \mathbf{dom} f, f(x) \leq t\}$. Show that f is a convex function if and only if $\mathbf{epi} f$ is a convex set.

Exercise 10 Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ and let $g : \mathbf{R} \rightarrow \mathbf{R}$ be defined as $g(t) = f(x + vt)$. Show that f is a convex function if and only if g is a convex function for all $x, v \in \mathbf{R}^n$. Here $\mathbf{dom} g = \{t \mid x + tv \in \mathbf{dom} f\}$.

Exercise 11 Show that $f(X) = -\ln \det X$ with $\mathbf{dom} f = \mathbf{S}_{++}^n$ is a convex function. Hint: Consider $g(t) = -\ln \det(X + Vt)$, where $X \in \mathbf{S}_{++}^n$ and $V \in \mathbf{S}^n$.